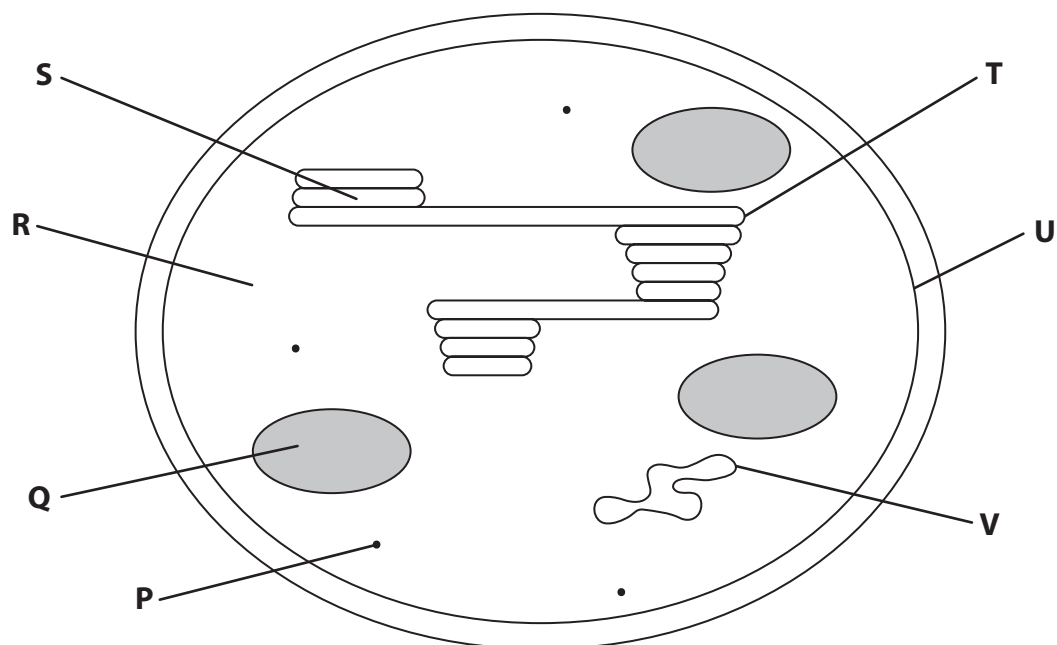


1 The structure of a chloroplast is related to its role in photosynthesis.

The diagram shows a chloroplast.



(a) (i) Which row in the table identifies the structures labelled **P**, **Q** and **V**?

(1)

	P	Q	V
☐ A	DNA	starch grain	ribosome
☐ B	starch grain	DNA	ribosome
☐ C	starch grain	ribosome	DNA
☐ D	ribosome	starch grain	DNA

(ii) Which structure contains GALP?

(1)

☐ **A Q**

☐ **B R**

☐ **C U**

☐ **D V**

(iii) The length of this chloroplast is $7.5\text{ }\mu\text{m}$.

Calculate the magnification of this diagram.

(1)

Answer

(iv) Structures **T** and **U** are membranes.

Compare and contrast the structure of these two membranes.

(3)

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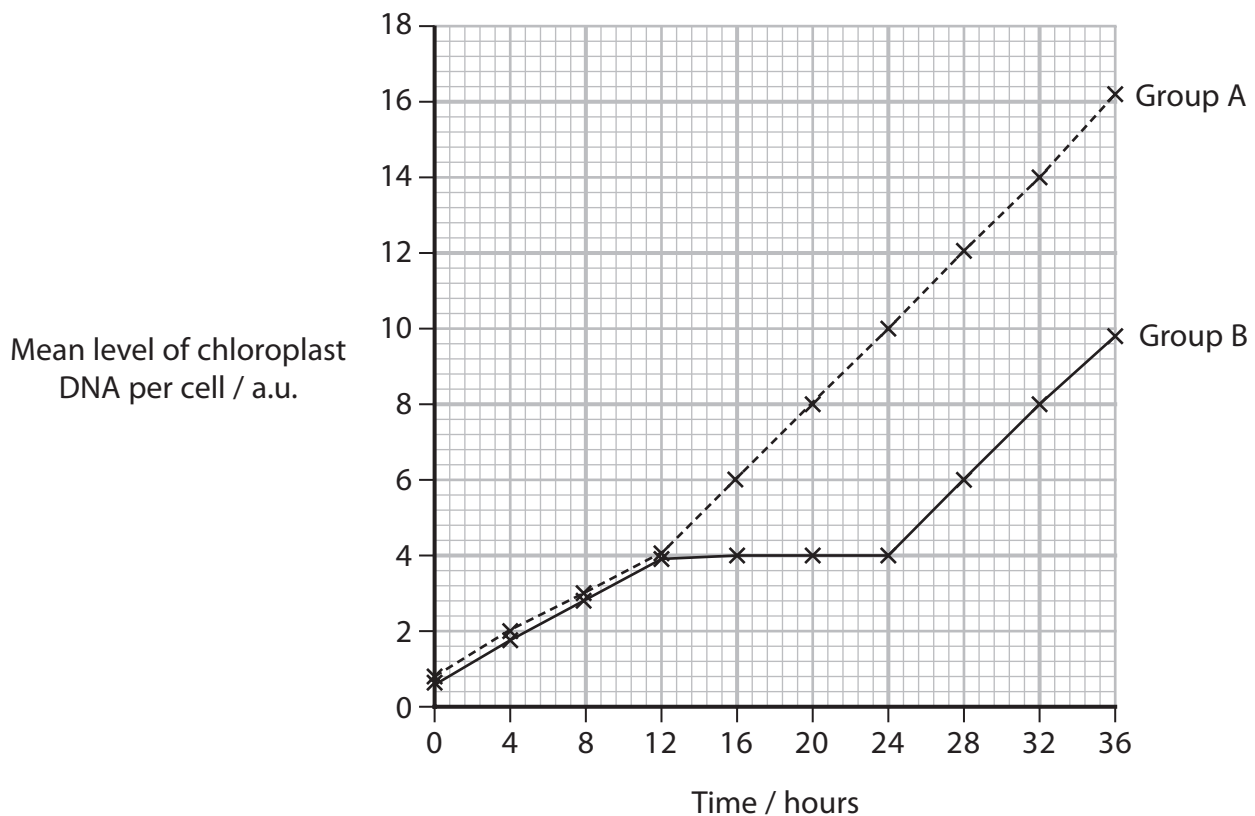
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(b) An investigation studied the effect of periods of light and dark on the levels of chloroplast DNA in one species of a single-celled organism.

One group of these organisms, group A, was exposed to 36 hours of continuous light.

The other group of these organisms, group B, was exposed to 12 hours of light, followed by 12 hours of darkness, followed by 12 hours of light.

The graph shows the results of this investigation.



These organisms divide by mitosis followed by cell division, during the dark.

Describe two conclusions that can be made about the replication of chloroplast DNA.

(2)

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(Total for Question 1 = 8 marks)